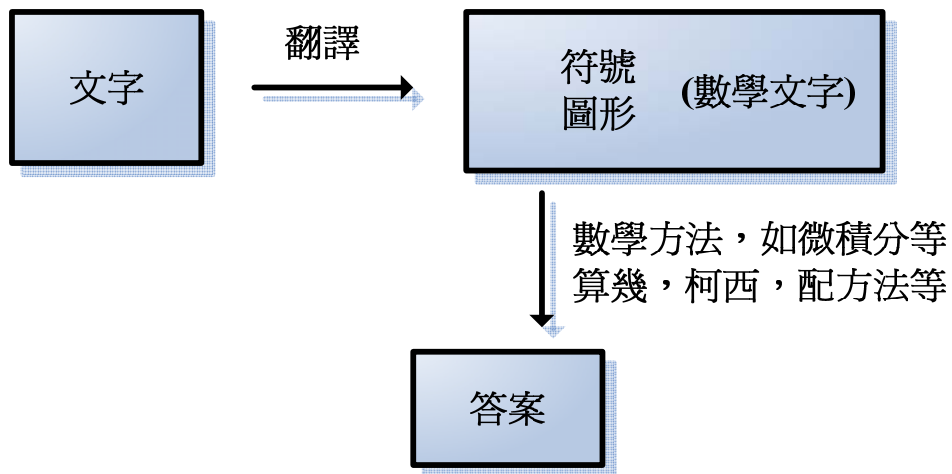


§4-7 Optimization

Homework : 4,10,17,24,29,33,35,51,57.



微積分方法：求最大值與最小值，可以用下列兩種方法來解。

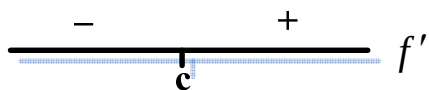
- (i) f is continuous on $[a, b]$.

比較

x	a	critical points	B
y			

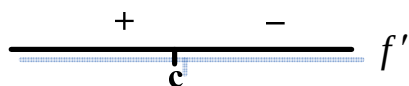
- (ii) f has exactly one critical point c in the interval I .

If



$\Rightarrow f(c)$ is an absolute minimum.

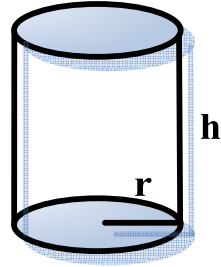
If :



$\Rightarrow f(c)$ is an absolute maximum.

Example 1 : A cylindrical can is to be made to hold 1 L of oil. Find the dimensions that will minimize the cost of the metal to manufacture the can?

Solution :



已知： $\pi r^2 h = 1000$

$$\begin{aligned} \text{Minimize } A &= 2\pi r^2 + 2\pi rh \\ &= 2\pi r^2 + \frac{2000}{r}, \quad 0 < r < \infty \end{aligned}$$

$$\begin{aligned} \frac{dA}{dr} &= 4\pi r - \frac{2000}{r^2} = \frac{4(\pi r^3 - 500)}{r^2} \\ \Rightarrow r &= \left(\frac{500}{\pi} \right)^{\frac{1}{3}} \end{aligned}$$

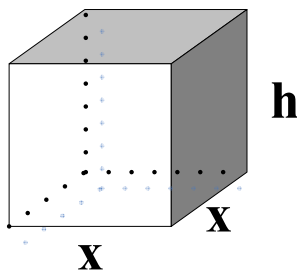
$$A \left[\left(\frac{500}{\pi} \right)^{\frac{1}{3}} \right] \text{ is a absolute minimum.}$$

$$\begin{array}{c} - \qquad \qquad + \\ \hline \qquad \qquad \left(\frac{500}{\pi} \right)^{\frac{1}{3}} \end{array} \Rightarrow$$

$$\Rightarrow h = \frac{1000}{\pi \left(\frac{500}{\pi} \right)^{\frac{2}{3}}} = 2\sqrt[3]{\frac{500}{\pi}} = 2r.$$

Example 2 : A box with a square base and open top must have a volume of $32,000 \text{ cm}^3$. Find the dimensions of the box that minimize the amount of material used.

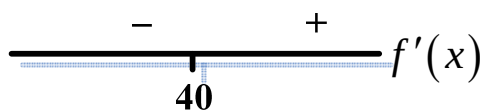
Solution :



已知 : $x^2 h = 32000$

$$\begin{aligned} \text{Minimize } x^2 + 4xh &= x^2 + \frac{128000}{x} \\ &=: f(x) \end{aligned}$$

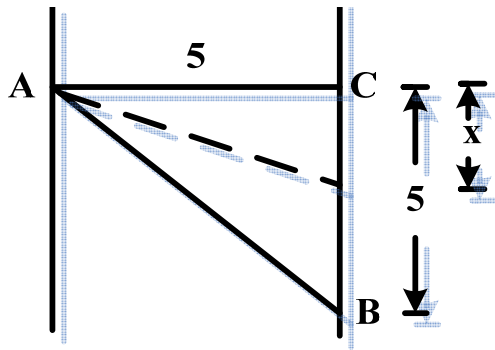
$$f'(x) = 2x - \frac{128000}{x^2} = \frac{2(x^3 - 64000)}{x^2}, \quad 0 < x < \infty$$



$$\Rightarrow x = 40 \quad h = 20.$$

Example 3 : A man launches his boat from point A on a bank of a straight river, 5 km wide, and wants to reach point B, 5 km downstream on the opposite bank, as quick as possible. If he can row 6 km/h and run 8 km/h, where should he land to reach B as soon as possible?

Solution :



Minimize

$$T(x) = \frac{\sqrt{25+x^2}}{6} + \frac{5-x}{8}, \quad 0 \leq x \leq 5$$

$$T'(x) = \frac{x}{6\sqrt{25+x^2}} - \frac{1}{8}$$

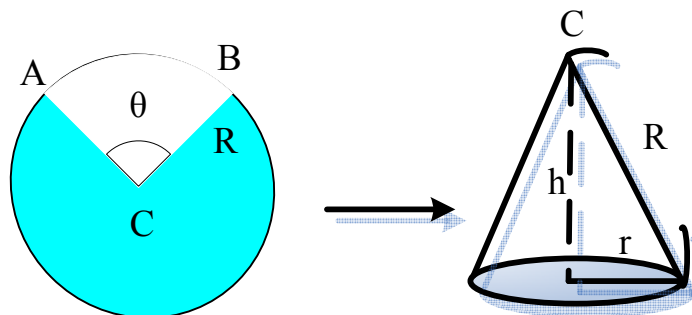
$$T'(x) = 0 \Rightarrow \frac{x}{6\sqrt{25+x^2}} = \frac{1}{8} \Rightarrow 16x^2 = 9x^2 + 225$$

$$\Rightarrow x = \frac{15}{\sqrt{7}} > 5$$

x	0	5
y	$\frac{5}{6} + \frac{5}{8}$	$\frac{5}{6}\sqrt{2}$

He should row directly to B .

Example 4 : A cone-shaped drinking cup is made from radius R by cutting out a sector and joining the edges CA and CB . Find the maximum capacity of such a cup.



Proof : $V(h) = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi (R^2 - h^2) h$

$$= \frac{\pi}{3} R^2 h - \frac{\pi}{3} h^3, \quad 0 < h < R$$

$$\frac{dV}{dh} = \frac{\pi}{3} (R^2 - 3h^2) = 0$$

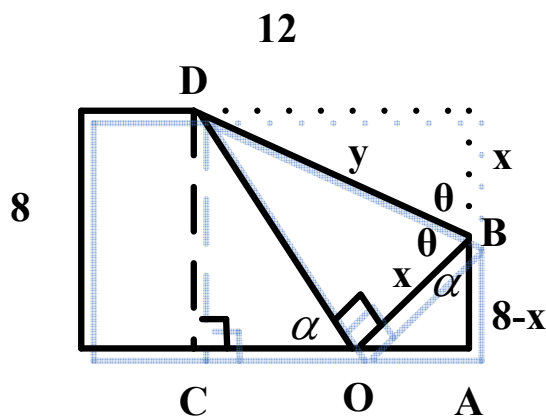
$$h = \frac{1}{\sqrt{3}} R$$

$$\begin{array}{c} + \qquad \qquad \qquad - \\ \hline \frac{1}{\sqrt{3}} R \end{array} \quad \frac{dV}{dh}$$

$$\Rightarrow V\left(\frac{1}{\sqrt{3}} R\right) = \frac{2\pi}{9\sqrt{3}} R^3 \quad \text{is the maximum.}$$

想想看：為達最大容量，角度 θ 要取多少？

Example 5: The upper right-hand corner of a piece of paper, 12 inch by 8 inch, as in the figure, is folded over to the bottom edge. How would you fold it, so as to minimize the length of the fold? In other words, how would you choose x to minimize y ?



Solution : $\triangle OCD \sim \triangle BAO$

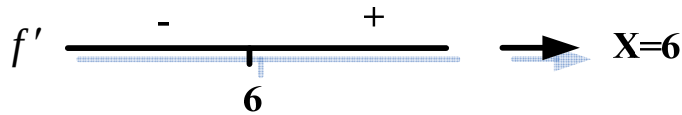
$$\Rightarrow \frac{8}{\sqrt{y^2 - x^2}} = \frac{4\sqrt{x-4}}{x}$$

$$\Rightarrow (y^2 - x^2)(x-4) = 4x^2$$

想想看：why $4 < x \leq 8$?

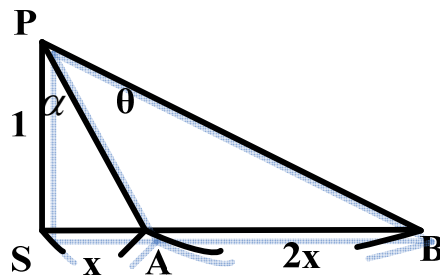
$$\Rightarrow y^2 = \frac{4x^2}{x-4} + x^2 = \frac{x^3}{x-4} =: f(x), \quad 4 < x \leq 8.$$

$$\Rightarrow f'(x) = \frac{2x^2(x-6)}{(x-4)^2} = 0 \Rightarrow x = 6$$



Example 6 : An observer stands at a point P, one unit away from a track. Two runners stand at the point S in the figure and run along the track. One runner runs three times as fast as other. Find the maximum value of the observer's angle of sight between runners?

Solution :



runner B runs three times as fast as runner A

$$\tan \theta = \tan(\theta + \alpha - \alpha) = \frac{\tan(\theta + \alpha) - \tan \alpha}{1 + \tan(\theta + \alpha) \tan \alpha}$$

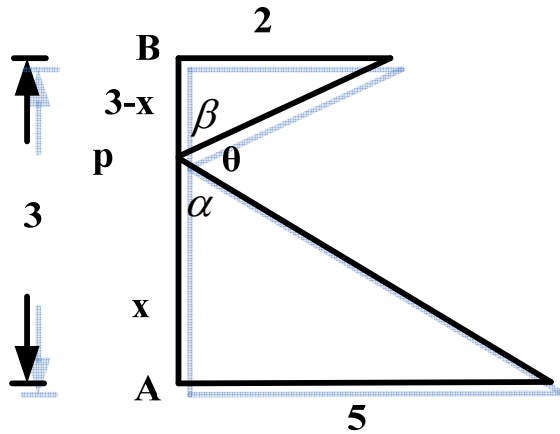
$$= \frac{3x - x}{1 + 3x^2} = \frac{2x}{1 + 3x^2} =: f(x), \quad 0 \leq x < \infty$$

$$f'(x) = \frac{2(1 - 3x^2)}{(1 + 3x^2)^2} = 0 \Rightarrow x = \frac{1}{\sqrt{3}} \quad (x \geq 0)$$

A horizontal number line with a tick mark at $\frac{1}{\sqrt{3}}$. To the left of this mark, the line is labeled with a plus sign (+). To the right, it is labeled with a minus sign (-). An arrow points from the number line to the equation $\Rightarrow \tan \theta = \frac{2 \frac{1}{\sqrt{3}}}{1 + 3 \left(\frac{1}{\sqrt{3}} \right)^2} = \frac{1}{\sqrt{3}}$.

$$\Rightarrow \theta = \frac{\pi}{6} \text{ is the max angle of sight.}$$

Example 7 : Where should the point p be chosen on the line segment AB so as to maximize the angle θ ?



Solution : $\tan \theta = \tan (\pi - (\alpha + \beta)) = -\tan (\alpha + \beta)$

$$= -\left(\frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} \right) = \frac{\frac{5}{x} + \frac{2}{3-x}}{\frac{5}{x} \cdot \frac{2}{3-x} - 1}$$

$$= \frac{3(5-x)}{x^2 - 3x + 10} =: f(x), \quad 0 \leq x \leq 5$$

$$f'(x) = \frac{3(x^2 - 10x + 5)}{(x^2 - 3x + 10)^2}$$

$$x = 5 - 2\sqrt{5} \quad (5 + 2\sqrt{5} > 3 \text{ 不合})$$

$$f' \quad \begin{array}{c} + \qquad \qquad - \\ \hline \end{array}$$

$$5 - 2\sqrt{5}$$

$$\Rightarrow x = 5 - 2\sqrt{5}$$

will maximize the angle θ